Tutorial Questions | Week 11

COSC2779 - Deep Learning

This tutorial reviews model interpretation: post-hoc explanation methods (Grad-CAM, LIME, SHAP, Integrated Gradients), interpretable-by-design models (concept bottlenecks), and explaining Transformers and vision–language models. Please try the questions before you join the session. The question below has a similar structure to the case study questions that you can expect in the end semester test.

1. You have been hired by a dermatology clinic network that uses a deep learning system to help triage skin lesions. The system has three parts:

   - a **CNN** that classifies dermoscopy images as *melanoma* or *benign*;
   - a **Transformer** text classifier that reads GP referral letters and flags *urgent* referrals;
   - a **vision–language model (VLM)** that is being trialled to draft a short report for each image.

   The clinic's governance committee will not approve deployment until the team can explain why the models make their predictions and show that those explanations can be trusted.

   (a) Why does interpretability matter particularly in this setting? Explain the difference between a **local** and a **global** explanation, giving one example of each for the melanoma classifier.

   (b) When the team applies **Grad-CAM** to melanoma predictions, the heatmaps often highlight a ruler and ink markings placed next to lesions, rather than the lesion itself. What does this reveal, and what would you do about it?

   (c) The classifier will be supplied by a vendor as an **API**: you can send images and receive probabilities, but have no access to weights or gradients. Can you still use Grad-CAM? Describe how **LIME** would explain a prediction in this setting.

   (d) To study an explanation more carefully, the team splits an image into two regions: the lesion **border** (A) and the lesion **colour pattern** (B). With the other region replaced by a neutral baseline, the model gives:

$v(\varnothing)$	$v(\{A\})$	$v(\{B\})$	$v(\{A, B\})$
0.1	0.5	0.3	0.9

   Compute the **Shapley values** ϕ_A and ϕ_B, and the **leave-one-out** scores for each region. Compare the two and comment.

   (e) In both LIME and SHAP, "removing" a region means replacing it with something (e.g. a black patch or a blur). Why can this choice make an explanation misleading, and how could you reduce the problem?

   (f) The dermatologists would prefer a model that reasons like they do using the **7-point checklist** (pigment network, streaks, blue-whitish veil, ...), and that they can correct. Propose an **interpretable-by-design** model. Describe its structure, how a clinician could intervene, and one limitation.

   (g) For the referral-letter Transformer, a colleague shows a map of the model's **attention weights** as the explanation of why a letter was flagged urgent. Is this justified? Suggest better alternatives for explaining a Transformer.

   (h) The team first tries **gradient** $\times$ **input** on the text model, but for letters the model is very confident about, all token scores are almost zero. Why? Explain how **Integrated Gradients** addresses this, including the role of the baseline.

   (i) When the VLM is asked "*Is this lesion suspicious? Explain your answer*", it gives a fluent, clinically worded justification. The governance committee asks: "Doesn't this give us explanations for free?" Respond, and explain how the type of access to the VLM (open weights vs a closed API) affects what you can do.

   (j) Several explanation methods now produce different heatmaps for the same images. Describe two tests the team could use to decide which explanations are **faithful** to the model.